

StructOptima Sample Report — G+1 Residential Building (12m × 9m)

1. Executive Summary

Total Floor Area	108.00 m²
Stories / Height	2 Stories / 6.0 m
RCC Concrete Volume	72.91 m³
Reinforcement Steel	2953 kg
Clay Modular Bricks	17,602 Nos
Masonry & Plaster Cement	158 Bags (50kg)
Masonry & Plaster Sand	38.48 Tonnes
Plaster Surface Area	703.1 m²
Total Estimated Cost	INR 1,596,882.24

2. Design Basis

Design Codes & Standards:

- IS 456:2000 (Code of Practice for Plain and Reinforced Concrete)

Material Properties:

- Concrete: M25 (Density: 25 kN/m³, $E = 5000\sqrt{f_{ck}}$)
- Steel: Fe415 (Density: 78.5 kN/m³, $E = 2 \times 10^5$ MPa)

Loading & Analysis:

- Design Load: User Input (Factored)
- Load Path: Tributary Area Method (Corner: 25%, Edge: 50%, Interior: 100% of bay)
- Beam Analysis: Simplified Simply Supported/Continuous assumptions.
($M = wL^2/8$ or $wL^2/10$, Deflection checked against $L/250$)

Column Design:

- Short Column Assumption ($L_e/r < 12$)
 - Sizing Criteria: Axial Stress Check ($P_u/Ag \leq 0.4 f_{ck}$)
- Note: Steel capacity ($0.67 f_y A_{sc}$) is reserved as safety margin.

Foundation Design:

- Isolated Square Footings
- Area Required = Load * 1.1 / SBC (1.1 factor for self-weight)
- Depth Check: Punching Shear at $d/2$ distance (Limit: $0.25\sqrt{f_{ck}}$)

Estimation Basis:

- Concrete Cost: INR 5000/m³
- Steel Cost: INR 60/kg
- Rebar Ratios: Columns 150kg/m³, Beams 120kg/m³, Footings 80kg/m³

3. Column Schedule & Quantities (IS 456 Cl. 26.5.3)

ID	Level	Loc (x,y)	Load (kN)	Size (mm)	Conc (m³)	Steel (kg)	Status
C1_L0	L0	(0.00, 0.00)	185.2	300x450	0.41	60.8	PASS
C1_L1	L1	(0.00, 0.00)	92.6	300x450	0.41	60.8	PASS
C2_L0	L0	(0.00, 4.50)	311.2	300x450	0.41	60.8	PASS
C2_L1	L1	(0.00, 4.50)	155.6	300x450	0.41	60.8	PASS

ID	Level	Loc (x,y)	Load (kN)	Size (mm)	Conc (m3)	Steel (kg)	Status
C3_L0	L0	(0.00, 9.00)	185.2	300x450	0.41	60.8	PASS
C3_L1	L1	(0.00, 9.00)	92.6	300x450	0.41	60.8	PASS
C4_L0	L0	(4.00, 0.00)	305.2	300x450	0.41	60.8	PASS
C4_L1	L1	(4.00, 0.00)	152.6	300x450	0.41	60.8	PASS
C5_L0	L0	(4.00, 4.50)	505.5	400x600	0.72	108.0	PASS
C5_L1	L1	(4.00, 4.50)	252.8	400x600	0.72	108.0	PASS
C6_L0	L0	(4.00, 9.00)	305.2	300x450	0.41	60.8	PASS
C6_L1	L1	(4.00, 9.00)	152.6	300x450	0.41	60.8	PASS
C7_L0	L0	(8.00, 0.00)	305.2	300x450	0.41	60.8	PASS
C7_L1	L1	(8.00, 0.00)	152.6	300x450	0.41	60.8	PASS
C8_L0	L0	(8.00, 4.50)	505.5	400x600	0.72	108.0	PASS
C8_L1	L1	(8.00, 4.50)	252.8	400x600	0.72	108.0	PASS
C9_L0	L0	(8.00, 9.00)	305.2	300x450	0.41	60.8	PASS
C9_L1	L1	(8.00, 9.00)	152.6	300x450	0.41	60.8	PASS
C10_L0	L0	(12.00, 0.00)	185.2	300x450	0.41	60.8	PASS
C10_L1	L1	(12.00, 0.00)	92.6	300x450	0.41	60.8	PASS
C11_L0	L0	(12.00, 4.50)	311.2	300x450	0.41	60.8	PASS
C11_L1	L1	(12.00, 4.50)	155.6	300x450	0.41	60.8	PASS
C12_L0	L0	(12.00, 9.00)	185.2	300x450	0.41	60.8	PASS
C12_L1	L1	(12.00, 9.00)	92.6	300x450	0.41	60.8	PASS

4. Reinforcement Schedule (IS 456 Cl. 26.5)

Col ID	Main Bars	Links (Stirrups)
C1_L0	8-16#	8# @ 150 c/c
C1_L1	8-16#	8# @ 200 c/c
C2_L0	8-16#	8# @ 150 c/c
C2_L1	8-16#	8# @ 200 c/c
C3_L0	8-16#	8# @ 150 c/c
C3_L1	8-16#	8# @ 200 c/c
C4_L0	8-16#	8# @ 150 c/c
C4_L1	8-16#	8# @ 200 c/c
C5_L0	10-16#	8# @ 150 c/c
C5_L1	10-16#	8# @ 200 c/c
C6_L0	8-16#	8# @ 150 c/c
C6_L1	8-16#	8# @ 200 c/c
C7_L0	8-16#	8# @ 150 c/c
C7_L1	8-16#	8# @ 200 c/c
C8_L0	10-16#	8# @ 150 c/c
C8_L1	10-16#	8# @ 200 c/c
C9_L0	8-16#	8# @ 150 c/c

Col ID	Main Bars	Links (Stirrups)
C9_L1	8-16#	8# @ 200 c/c
C10_L0	8-16#	8# @ 150 c/c
C10_L1	8-16#	8# @ 200 c/c
C11_L0	8-16#	8# @ 150 c/c
C11_L1	8-16#	8# @ 200 c/c
C12_L0	8-16#	8# @ 150 c/c
C12_L1	8-16#	8# @ 200 c/c

5. Foundation Schedule & Quantities (IS 456 Cl. 34)

Col ID	Load (kN)	Size (m)	Depth (mm)	Conc (m3)	Steel (kg)
N/A	-	-	-	-	-

6. Beam Schedule & Quantities (IS 456 Cl. 26.5.1)

ID	Size (mm)	Span (m)	Conc (m3)	Steel (kg)	Top Steel	Bot Steel	Stirrups
B_H_1	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_2	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_3	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_4	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_5	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_6	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_7	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_8	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_H_9	230x350	4.00	0.32	38.6	2-12#	5-20#	2L-8# @ 230 c/c
B_V_10	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c
B_V_11	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c
B_V_12	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c
B_V_13	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c
B_V_14	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c
B_V_15	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c
B_V_16	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c
B_V_17	230x400	4.00	0.37	44.2	2-12#	5-20#	2L-8# @ 230 c/c

7. Slab Schedule & Quantities

ID	Thk (mm)	Area (m2)	Conc (m3)	Steel (kg)	Main Mesh	Dist Mesh
S1	185	16.0	2.96	266.4	10# @ 300 c/c	10# @ 300 c/c
S2	185	16.0	2.96	266.4	10# @ 300 c/c	10# @ 300 c/c
S3	185	16.0	2.96	266.4	10# @ 300 c/c	10# @ 300 c/c
S4	185	16.0	2.96	266.4	10# @ 300 c/c	10# @ 300 c/c
S5	185	16.0	2.96	266.4	10# @ 300 c/c	10# @ 300 c/c
S6	185	16.0	2.96	266.4	10# @ 300 c/c	10# @ 300 c/c

8. Brickwork Schedule & Material Takeoff (IS 1077, IS 2250)

Modular clay bricks 200×200×100 mm with mortar (190×190×90 mm solid core, 250 bricks/m³ per IS 1077). Mortar 1:4 mix proportion with 1.33 dry volume factor per IS 2250. W:C ratio = 0.50.

Wall ID	Span / Cols	Len (m)	Ht (m)	Thk	Gross (m³)	Bricks (Nos)	Brick Vol	Wet Mor	Dry Mor	Cem Bags	Cem (kg)	Sand (T)	Water (L)
W_H1	C1_L0→C4_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H2	C4_L0→C7_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H3	C7_L0→C10_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H4	C2_L0→C5_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H5	C5_L0→C8_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H6	C8_L0→C11_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H7	C3_L0→C6_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H8	C6_L0→C9_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H9	C9_L0→C12_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_V10	C1_L0→C2_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
W_V11	C2_L0→C3_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
W_V12	C4_L0→C5_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
W_V13	C5_L0→C6_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
W_V14	C7_L0→C8_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
W_V15	C8_L0→C9_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
W_V16	C10_L0→C11_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
W_V17	C11_L0→C12_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
TOTAL	All Walls	-	-	-	71.28	17,822	57.90	13.38	17.79	103	5,124	22.77	2,562

9. Plastering Schedule & Quantities (IS 1661, IS 2402, IS 1200)

External wall plastering: 15mm thick in 1:4 cement:sand mix (IS 2402). Internal wall plastering: 12mm thick in 1:6 mix (IS 1661). Ceiling soffit plastering: 6mm thick in 1:3 mix per IS 1661 Cl. 12.3. Dry volume factor = 1.33.

Surface ID	Location & Mix Spec	Area (m²)	Thk (mm)	Mix	Wet Vol (m³)	Dry Vol (m³)	Cem (Bags)	Cem (kg)	Sand (T)	Water (L)
W_H1_EXT	Ext Render (15mm 1:4)	19.8	15	1:4	0.297	0.395	3	113.8	0.51	57
W_H1_INT	Int Wall (12mm 1:6)	19.8	12	1:6	0.238	0.316	2	65.0	0.43	33
W_H2_EXT	Ext Render (15mm 1:4)	19.8	15	1:4	0.297	0.395	3	113.8	0.51	57
W_H2_INT	Int Wall (12mm 1:6)	19.8	12	1:6	0.238	0.316	2	65.0	0.43	33
W_H3_EXT	Ext Render (15mm 1:4)	19.8	15	1:4	0.297	0.395	3	113.8	0.51	57
W_H3_INT	Int Wall (12mm 1:6)	19.8	12	1:6	0.238	0.316	2	65.0	0.43	33
W_H4	Int Part (2-Face 12mm 1:6)	39.6	12	1:6	0.475	0.632	3	130.0	0.87	65
W_H5	Int Part (2-Face 12mm 1:6)	39.6	12	1:6	0.475	0.632	3	130.0	0.87	65
W_H6	Int Part (2-Face 12mm 1:6)	39.6	12	1:6	0.475	0.632	3	130.0	0.87	65
W_H7_EXT	Ext Render (15mm 1:4)	19.8	15	1:4	0.297	0.395	3	113.8	0.51	57
W_H7_INT	Int Wall (12mm 1:6)	19.8	12	1:6	0.238	0.316	2	65.0	0.43	33
W_H8_EXT	Ext Render (15mm 1:4)	19.8	15	1:4	0.297	0.395	3	113.8	0.51	57
W_H8_INT	Int Wall (12mm 1:6)	19.8	12	1:6	0.238	0.316	2	65.0	0.43	33

Surface ID	Location & Mix Spec	Area (m²)	Thk (mm)	Mix	Wet Vol (m³)	Dry Vol (m³)	Cem (Bags)	Cem (kg)	Sand (T)	Water (L)
W_H9_EXT	Ext Render (15mm 1:4)	19.8	15	1:4	0.297	0.395	3	113.8	0.51	57
W_H9_INT	Int Wall (12mm 1:6)	19.8	12	1:6	0.238	0.316	2	65.0	0.43	33
W_V10_EXT	Ext Render (15mm 1:4)	22.3	15	1:4	0.334	0.444	3	128.0	0.57	64
W_V10_INT	Int Wall (12mm 1:6)	22.3	12	1:6	0.267	0.356	2	73.1	0.49	37
W_V11_EXT	Ext Render (15mm 1:4)	22.3	15	1:4	0.334	0.444	3	128.0	0.57	64
W_V11_INT	Int Wall (12mm 1:6)	22.3	12	1:6	0.267	0.356	2	73.1	0.49	37
W_V12	Int Part (2-Face 12mm 1:6)	44.6	12	1:6	0.535	0.711	3	146.3	0.98	73
W_V13	Int Part (2-Face 12mm 1:6)	44.6	12	1:6	0.535	0.711	3	146.3	0.98	73
W_V14	Int Part (2-Face 12mm 1:6)	44.6	12	1:6	0.535	0.711	3	146.3	0.98	73
W_V15	Int Part (2-Face 12mm 1:6)	44.6	12	1:6	0.535	0.711	3	146.3	0.98	73
W_V16_EXT	Ext Render (15mm 1:4)	22.3	15	1:4	0.334	0.444	3	128.0	0.57	64
W_V16_INT	Int Wall (12mm 1:6)	22.3	12	1:6	0.267	0.356	2	73.1	0.49	37
W_V17_EXT	Ext Render (15mm 1:4)	22.3	15	1:4	0.334	0.444	3	128.0	0.57	64
W_V17_INT	Int Wall (12mm 1:6)	22.3	12	1:6	0.267	0.356	2	73.1	0.49	37
Ceiling Soffit	Slab Bottom (6mm 1:3)	216.0	6	1:3	1.296	1.724	13	620.5	2.07	310
TOTAL	All Plaster Surfaces	928.8	-	-	10.473	13.929	70	3,473	18.43	1,736

10. Painting & Surface Finishes Schedule (NBC 2016 SP7)

Surface preparation & coatings per NBC 2016 SP7: Water-thinnable primer (1 coat @ 8 m²/L), polymer wall putty (2 coats @ 1.5 kg/m²), interior acrylic emulsion (2 coats @ 12 m²/L), exterior weather-guard acrylic emulsion (2 coats @ 10 m²/L).

Surface ID	Location & Specification	Finish Area (m²)	Primer 1-Ct (L)	Putty 2-Ct (kg)	Int Emulsion 2-Ct (L)	Ext Weather 2-Ct (L)
W_H1_EXT	External Weather Guard	19.8	2.5	—	—	4.0
W_H1_INT	Internal Acrylic Emulsion	19.8	2.5	29.7	3.3	—
W_H2_EXT	External Weather Guard	19.8	2.5	—	—	4.0
W_H2_INT	Internal Acrylic Emulsion	19.8	2.5	29.7	3.3	—
W_H3_EXT	External Weather Guard	19.8	2.5	—	—	4.0
W_H3_INT	Internal Acrylic Emulsion	19.8	2.5	29.7	3.3	—
W_H4	Internal Partition Emulsion	39.6	5.0	59.4	6.6	—
W_H5	Internal Partition Emulsion	39.6	5.0	59.4	6.6	—
W_H6	Internal Partition Emulsion	39.6	5.0	59.4	6.6	—
W_H7_EXT	External Weather Guard	19.8	2.5	—	—	4.0
W_H7_INT	Internal Acrylic Emulsion	19.8	2.5	29.7	3.3	—
W_H8_EXT	External Weather Guard	19.8	2.5	—	—	4.0
W_H8_INT	Internal Acrylic Emulsion	19.8	2.5	29.7	3.3	—
W_H9_EXT	External Weather Guard	19.8	2.5	—	—	4.0
W_H9_INT	Internal Acrylic Emulsion	19.8	2.5	29.7	3.3	—
W_V10_EXT	External Weather Guard	22.3	2.8	—	—	4.5
W_V10_INT	Internal Acrylic Emulsion	22.3	2.8	33.4	3.7	—
W_V11_EXT	External Weather Guard	22.3	2.8	—	—	4.5
W_V11_INT	Internal Acrylic Emulsion	22.3	2.8	33.4	3.7	—
W_V12	Internal Partition Emulsion	44.6	5.6	66.8	7.4	—

Surface ID	Location & Specification	Finish Area (m²)	Primer 1-Ct (L)	Putty 2-Ct (kg)	Int Emulsion 2-Ct (L)	Ext Weather 2-Ct (L)
W_V13	Internal Partition Emulsion	44.6	5.6	66.8	7.4	—
W_V14	Internal Partition Emulsion	44.6	5.6	66.8	7.4	—
W_V15	Internal Partition Emulsion	44.6	5.6	66.8	7.4	—
W_V16_EXT	External Weather Guard	22.3	2.8	—	—	4.5
W_V16_INT	Internal Acrylic Emulsion	22.3	2.8	33.4	3.7	—
W_V17_EXT	External Weather Guard	22.3	2.8	—	—	4.5
W_V17_INT	Internal Acrylic Emulsion	22.3	2.8	33.4	3.7	—
Ceiling Soffit	Slab Underside (White Emulsion)	216.0	27.0	324.0	36.0	—
TOTAL	All Surfaces (Walls + Ceilings)	928.8	116.1	1,081.4	120.1	41.6

11. Consolidated Masonry & Finishes BOQ & Cost (IS 1905, CPWD DSR)

11.1 Comprehensive Bill of Materials

Item / Material Description	Specification / Mix Proportion	Building Total	Unit	IS Standard
Total Wall Gross Volume	L × t × Hv (clear story height)	71.28	m³	IS 1905:1987
Clay Modular Building Bricks	Standard modular 20×20×10 cm	17,822	Nos	IS 1077:1992
Net Solid Brickwork Volume	Actual brick body volume (excl. mortar)	57.90	m³	IS 1077:1992
Wet Brickwork Mortar Volume	Gross Wall Volume – Solid Brick Vol	13.38	m³	IS 2250:1981
Dry Brickwork Mortar Volume	Wet Volume × 1.33 dry factor	17.79	m³	IS 2250:1981
Brickwork Cement	1:4 Mix Proportion (OPC / PPC)	103 (5,124 kg)	Bags (50kg)	IS 269:2015
Brickwork Sand	Coarse sand, dry density 1600 kg/m³	22.77	Tonnes	IS 383 Zone II
Brickwork Water	Water:Cement ratio = 0.50	2,562	Litres	IS 456 Cl 5.4
Wall Plastering Surface Area	Internal (12mm 1:6) + External (15mm 1:4)	712.8	m²	IS 1200 Pt 12
Ceiling Soffit Plaster Area	Soffit bottom of slabs (6mm 1:3 mix)	216.0	m²	IS 1661:1972
Total Plaster Surface Area	Combined wall + ceiling surface	928.8	m²	IS 1200 Pt 12
Plastering Cement	Combined wall & ceiling plaster	70 (3,473 kg)	Bags (50kg)	IS 1661 / IS 2402
Plastering Sand	Clean plaster sand (Zone III/IV)	18.43	Tonnes	IS 383:2016
Plastering Water	Water:Cement ratio = 0.50	1,736	Litres	IS 456 Cl 5.4
Surface Primer	Water-thinnable primer @ 8 m²/L (1 coat)	116.1	Litres	NBC 2016 SP7
Polymer Wall Putty	2 coats (1–2 mm) @ 1.5 kg/m²	1,838.7	kg	NBC 2016 SP7
Interior Acrylic Emulsion	2 coats (30–40 microns) @ 12 m²/L	120.2	Litres	NBC 2016 SP7
Exterior Weather Emulsion	2 coats (40–60 microns) @ 10 m²/L	41.6	Litres	NBC 2016 SP7
GRAND TOTAL CEMENT (Mortar + Plaster)	All masonry + plastering works combined	173 bags (8,596 kg)	Bags	IS 269:2015
GRAND TOTAL SAND (Mortar + Plaster)	All masonry + plastering sand combined	41.20	Tonnes	IS 383:2016
GRAND TOTAL WATER (Mortar + Plaster)	All masonry + plastering water combined	4,298	Litres	IS 456 Cl 5.4

11.2 Trade-wise Cost Takeoff

Trade / Material Component	Quantity	Rate (INR)	Subtotal (INR)
Clay Modular Bricks (IS 1077)	17,822 Nos	■9 / brick	■160,398.00

Brickwork Mortar Cement	103 Bags	■380 / bag	■39,140.00
Brickwork Mortar Sand	22.77 Tonnes	■1,500 / Tonne	■34,157.78
Plastering Work (12/15/6mm)	928.8 m²	■280 / m²	■260,064.00
Surface Finishes (Primer + Putty + Paint)	928.8 m²	■120 / m²	■111,456.00
TOTAL MASONRY, PLASTER & FINISHES			■605,215.78

12. Structural Layout Plan

Column Coordinates: C1_L0: (0.00, 0.00) C2_L0: (0.00, 4.50) C3_L0: (0.00, 9.00) C4_L0: (4.00, 0.00) C5_L0: (4.00, 4.50) C6_L0: (4.00, 9.00) C7_L0: (8.00, 0.00) C8_L0: (8.00, 4.50) C9_L0: (8.00, 9.00) C10_L0: (12.00, 0.00) C11_L0: (12.00, 4.50) C12_L0: (12.00, 9.00) C1_L1: (0.00, 0.00) C2_L1: (0.00, 4.50) C3_L1: (0.00, 9.00) C4_L1: (4.00, 0.00) C5_L1: (4.00, 4.50) C6_L1: (4.00, 9.00) C7_L1: (8.00, 0.00) C8_L1: (8.00, 4.50) C9_L1: (8.00, 9.00) C10_L1: (12.00, 0.00) C11_L1: (12.00, 4.50) C12_L1: (12.00, 9.00)

13. Integrated Steel Breakdown

Bar Diameter	Total Weight (kg)
8 mm	250.8
10 mm	1717.4
16 mm	984.9
TOTAL	2953.1

14. Structural Audit Report

Member	Check	Design Val	Limit	Status
C1_L0	Column Axial Capacity	185.2	1781.2	PASS
C2_L0	Column Axial Capacity	311.2	1781.2	PASS
C3_L0	Column Axial Capacity	185.2	1781.2	PASS
C4_L0	Column Axial Capacity	305.2	1781.2	PASS
C5_L0	Column Axial Capacity	505.5	2938.9	PASS
C6_L0	Column Axial Capacity	305.2	1781.2	PASS
C7_L0	Column Axial Capacity	305.2	1781.2	PASS
C8_L0	Column Axial Capacity	505.5	2938.9	PASS
C9_L0	Column Axial Capacity	305.2	1781.2	PASS
C10_L0	Column Axial Capacity	185.2	1781.2	PASS
C11_L0	Column Axial Capacity	311.2	1781.2	PASS
C12_L0	Column Axial Capacity	185.2	1781.2	PASS
C1_L1	Column Axial Capacity	92.6	1781.2	PASS
C2_L1	Column Axial Capacity	155.6	1781.2	PASS
C3_L1	Column Axial Capacity	92.6	1781.2	PASS
C4_L1	Column Axial Capacity	152.6	1781.2	PASS
C5_L1	Column Axial Capacity	252.8	2938.9	PASS
C6_L1	Column Axial Capacity	152.6	1781.2	PASS
C7_L1	Column Axial Capacity	152.6	1781.2	PASS
C8_L1	Column Axial Capacity	252.8	2938.9	PASS
C9_L1	Column Axial Capacity	152.6	1781.2	PASS

Member	Check	Design Val	Limit	Status
C10_L1	Column Axial Capacity	92.6	1781.2	PASS
C11_L1	Column Axial Capacity	155.6	1781.2	PASS
C12_L1	Column Axial Capacity	92.6	1781.2	PASS
F-C1_L0	Bearing Pressure	60.4	200.0	PASS
F-C1_L0	Punching Shear	0.3	1.2	PASS
F-C2_L0	Bearing Pressure	101.4	200.0	PASS
F-C2_L0	Punching Shear	0.6	1.2	PASS
F-C3_L0	Bearing Pressure	60.4	200.0	PASS
F-C3_L0	Punching Shear	0.3	1.2	PASS
F-C4_L0	Bearing Pressure	99.5	200.0	PASS
F-C4_L0	Punching Shear	0.6	1.2	PASS
F-C5_L0	Bearing Pressure	92.7	200.0	PASS
F-C5_L0	Punching Shear	0.6	1.2	PASS
F-C6_L0	Bearing Pressure	99.5	200.0	PASS
F-C6_L0	Punching Shear	0.6	1.2	PASS
F-C7_L0	Bearing Pressure	99.5	200.0	PASS
F-C7_L0	Punching Shear	0.6	1.2	PASS
F-C8_L0	Bearing Pressure	92.7	200.0	PASS
F-C8_L0	Punching Shear	0.6	1.2	PASS
F-C9_L0	Bearing Pressure	99.5	200.0	PASS
F-C9_L0	Punching Shear	0.6	1.2	PASS
F-C10_L0	Bearing Pressure	60.4	200.0	PASS
F-C10_L0	Punching Shear	0.3	1.2	PASS
F-C11_L0	Bearing Pressure	101.4	200.0	PASS
F-C11_L0	Punching Shear	0.6	1.2	PASS
F-C12_L0	Bearing Pressure	60.4	200.0	PASS
F-C12_L0	Punching Shear	0.3	1.2	PASS
B_H_1_L0	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_2_L0	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_3_L0	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_4_L0	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_5_L0	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_6_L0	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_7_L0	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_8_L0	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_9_L0	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_V_10_L0	Beam Deflection (Serviceability)	7.8	18.0	PASS
B_V_11_L0	Beam Deflection (Serviceability)	7.8	18.0	PASS
B_V_12_L0	Beam Deflection (Serviceability)	7.8	18.0	PASS
B_V_13_L0	Beam Deflection (Serviceability)	7.8	18.0	PASS
B_V_14_L0	Beam Deflection (Serviceability)	7.8	18.0	PASS

Member	Check	Design Val	Limit	Status
B_V_15_L	Beam Deflection (Serviceability)	7.8	18.0	PASS
B_V_16_L	Beam Deflection (Serviceability)	7.8	18.0	PASS
B_V_17_L	Beam Deflection (Serviceability)	7.8	18.0	PASS
B_H_1_L	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_2_L	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_3_L	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_4_L	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_5_L	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_6_L	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_7_L	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_8_L	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_H_9_L	Beam Deflection (Serviceability)	7.1	16.0	PASS
B_V_10_L	Beam Deflection (Serviceability)	7.8	18.0	PASS
B_V_11_L	Beam Deflection (Serviceability)	7.8	18.0	PASS
B_V_12_L	Beam Deflection (Serviceability)	7.8	18.0	PASS
B_V_13_L	Beam Deflection (Serviceability)	7.8	18.0	PASS
B_V_14_L	Beam Deflection (Serviceability)	7.8	18.0	PASS
B_V_15_L	Beam Deflection (Serviceability)	7.8	18.0	PASS
B_V_16_L	Beam Deflection (Serviceability)	7.8	18.0	PASS
B_V_17_L	Beam Deflection (Serviceability)	7.8	18.0	PASS

15. Sustainability Audit

Design Strategy: Standard (OPC)

Material	Quantity	Emission Factor	Total CO2e (kg)
Concrete	72.91 m3	300 kg/m3	21873.0
Steel	2953 kg	1.85 kg/kg	5463.3
TOTAL	-	-	27336.3

Wait! You can reduce your Carbon Footprint.

Refining the design to use **Fly Ash based Concrete (PPC)** instead of OPC can reduce concrete emissions by approximately 15-20%.

Switch "Use Green Concrete" ON in the dashboard to see savings. (Note: PPC is slower setting and has lower early strength, use only if confirmed by RMC supplier).

16. Serviceability Checks (Deflection)

Beam ID	Description	Actual (mm)	Limit (mm)	Status
B_H_1_L0	Span/250 Check	7.143	16.00	PASS
B_H_2_L0	Span/250 Check	7.143	16.00	PASS
B_H_3_L0	Span/250 Check	7.143	16.00	PASS
B_H_4_L0	Span/250 Check	7.143	16.00	PASS
B_H_5_L0	Span/250 Check	7.143	16.00	PASS
B_H_6_L0	Span/250 Check	7.143	16.00	PASS
B_H_7_L0	Span/250 Check	7.143	16.00	PASS
B_H_8_L0	Span/250 Check	7.143	16.00	PASS
B_H_9_L0	Span/250 Check	7.143	16.00	PASS
B_V_10_L0	Span/250 Check	7.765	18.00	PASS
B_V_11_L0	Span/250 Check	7.765	18.00	PASS
B_V_12_L0	Span/250 Check	7.765	18.00	PASS
B_V_13_L0	Span/250 Check	7.765	18.00	PASS
B_V_14_L0	Span/250 Check	7.765	18.00	PASS
B_V_15_L0	Span/250 Check	7.765	18.00	PASS
B_V_16_L0	Span/250 Check	7.765	18.00	PASS
B_V_17_L0	Span/250 Check	7.765	18.00	PASS
B_H_1_L1	Span/250 Check	7.143	16.00	PASS
B_H_2_L1	Span/250 Check	7.143	16.00	PASS
B_H_3_L1	Span/250 Check	7.143	16.00	PASS
B_H_4_L1	Span/250 Check	7.143	16.00	PASS
B_H_5_L1	Span/250 Check	7.143	16.00	PASS
B_H_6_L1	Span/250 Check	7.143	16.00	PASS
B_H_7_L1	Span/250 Check	7.143	16.00	PASS
B_H_8_L1	Span/250 Check	7.143	16.00	PASS
B_H_9_L1	Span/250 Check	7.143	16.00	PASS
B_V_10_L1	Span/250 Check	7.765	18.00	PASS
B_V_11_L1	Span/250 Check	7.765	18.00	PASS
B_V_12_L1	Span/250 Check	7.765	18.00	PASS
B_V_13_L1	Span/250 Check	7.765	18.00	PASS
...	...	...	...	...

17. Staircase Design Schedule (Dog-Legged)

No staircase selected for design.

20. Concrete Material Breakdown (IS 10262:2019)

Material	Per m ³	Total
Wet Concrete Volume		72.91 m ³
Cement (50kg bags)	7.6 bags	554.1 bags
Fine Aggregate (Sand)	0.419 m ³	30.53 m ³
Coarse Aggregate (20mm)	0.821 m ³	59.84 m ³
Water	171 kg	12468 kg

Water:Cement Ratio	0.45	—
Nominal Mix Ratio	1:1.76:3.13	—

Note: Indicative quantities per IS 10262:2019. Add 3-5% wastage for actual procurement.

23. Code Compliance Audit

Check	Status	Notes
Column Axial Capacity	PASS	Load: 185.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 311.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 185.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 305.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 505.5 vs Cap: 2938.9 kN
Column Axial Capacity	PASS	Load: 305.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 305.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 505.5 vs Cap: 2938.9 kN
Column Axial Capacity	PASS	Load: 305.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 185.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 311.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 185.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 92.6 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 155.6 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 92.6 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 152.6 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 252.8 vs Cap: 2938.9 kN
Column Axial Capacity	PASS	Load: 152.6 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 152.6 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 252.8 vs Cap: 2938.9 kN
Column Axial Capacity	PASS	Load: 152.6 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 92.6 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 155.6 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 92.6 vs Cap: 1781.2 kN
Bearing Pressure	PASS	Press: 60.4 <= 200.0
Punching Shear	PASS	Shear Stress: 0.34 <= 1.25 N/mm ²
Bearing Pressure	PASS	Press: 101.4 <= 200.0
Punching Shear	PASS	Shear Stress: 0.58 <= 1.25 N/mm ²
Bearing Pressure	PASS	Press: 60.4 <= 200.0
Punching Shear	PASS	Shear Stress: 0.34 <= 1.25 N/mm ²
Bearing Pressure	PASS	Press: 99.5 <= 200.0
Punching Shear	PASS	Shear Stress: 0.57 <= 1.25 N/mm ²
Bearing Pressure	PASS	Press: 92.7 <= 200.0
Punching Shear	PASS	Shear Stress: 0.59 <= 1.25 N/mm ²
Bearing Pressure	PASS	Press: 99.5 <= 200.0
Punching Shear	PASS	Shear Stress: 0.57 <= 1.25 N/mm ²

Check	Status	Notes
Bearing Pressure	PASS	Press: 99.5 <= 200.0
Punching Shear	PASS	Shear Stress: 0.57 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 92.7 <= 200.0
Punching Shear	PASS	Shear Stress: 0.59 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 99.5 <= 200.0
Punching Shear	PASS	Shear Stress: 0.57 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 60.4 <= 200.0
Punching Shear	PASS	Shear Stress: 0.34 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 101.4 <= 200.0
Punching Shear	PASS	Shear Stress: 0.58 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 60.4 <= 200.0
Punching Shear	PASS	Shear Stress: 0.34 <= 1.25 N/mm2
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.143mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm

Check	Status	Notes
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.765mm vs Limit 18.00mm

24. Professional Disclaimer

IMPORTANT NOTICE - READ CAREFULLY

This structural design report is generated by automated software for **PRELIMINARY DESIGN AND EDUCATIONAL PURPOSES ONLY**.

Key Limitations:

- The software uses simplified analysis methods (tributary area, short column assumptions).
- Seismic forces (IS 1893) and Wind loads (IS 875 Pt 3) have been incorporated for basic checks, but require full dynamic analysis for high-rise buildings.
- Site-specific soil conditions, groundwater, and other geotechnical factors are not evaluated.
- Local building codes may have additional requirements beyond IS 456:2000.

Required Actions Before Construction:

- All designs **MUST** be reviewed and sealed by a licensed Professional Engineer (PE).
- Independent structural analysis should be performed using appropriate software.
- Geotechnical investigation and foundation recommendations are required.
- All applicable permits and approvals must be obtained.

Liability Disclaimer:

The developers, operators, and distributors of this software assume **NO LIABILITY** for any damages, losses, or injuries arising from the use of designs produced by this tool. Use of this software constitutes acceptance of this disclaimer.

Generated by StructOptima. This report is a preliminary design aid compliant with IS 456:2000, IS 1893:2016, IS 13920:2016.

All designs must be verified by a licensed Structural Engineer prior to construction.

Safety Factors: $\gamma_c = 1.5$ (concrete), $\gamma_s = 1.15$ (steel) | Load Factor: 1.5 (DL + LL)

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